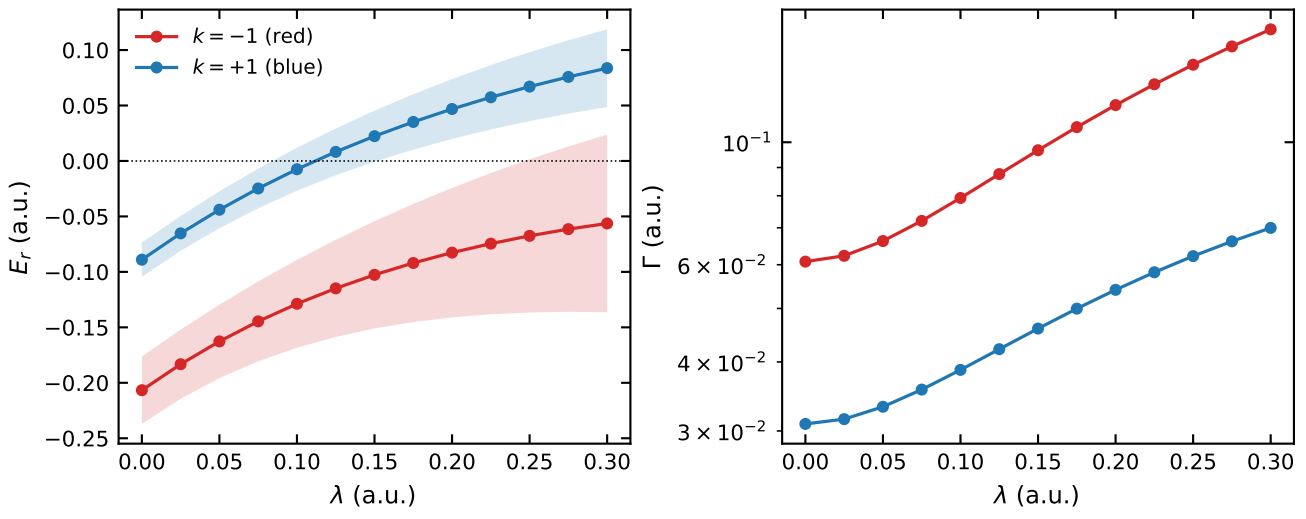
$k = -1$ (red)
$k = +1$ (blue)
E_r (a.u.)
λ (a.u.)
Γ (a.u.)
λ (a.u.)